An Accuracy and Calibration Benchmark of scGPT Pipelines and Matched Classical Models for Cell Type Annotation

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Abstract

Pretrained single-cell models are intended to provide transferable representations, but their advantage over supervised baselines and the reliability of probabilities produced by downstream annotation pipelines remain uncertain. We compared official frozen scGPT cell embeddings with normalized expression of the same training-selected and vocabulary-matched genes. Each representation was paired with Logistic Regression, Random Forest, and XGBoost across six public atlases containing 1,405,933 cells. We used training-only feature selection, five donor-held-out folds in 199 Indonesian donors, cell and donor bootstrap analyses, five calibration measures, a 10–500-label-per-class scarcity experiment, and held-out-donor temperature scaling. The strongest matched-expression pipeline achieved higher observed Macro F1 than the strongest frozen-scGPT pipeline in all six atlases. In donor-held-out evaluation, pooled Macro F1 was 0.956 for matched expression and 0.934 for scGPT embeddings; the paired difference was 0.022 with a donor-bootstrap 95% interval of 0.021–0.024. Matched expression remained stronger throughout the tested label range. Temperature scaling reduced calibration error and negative log-likelihood without changing class predictions. These results establish a pipeline-specific advantage for matched classical models under the tested supervised within-atlas conditions, while leaving zero-shot, external-atlas, rare-cell, and fine-grained transfer settings unresolved.

Keywords: single-cell RNA sequencing; scGPT; cell type annotation; calibration; donor-held-out validation; benchmarking

1 Introduction

Single-cell RNA sequencing (scRNA-seq) has changed the scale at which cellular heterogeneity can be studied. Droplet technologies now support measurements from hundreds of thousands of cells, and coordinated projects such as the Human Cell Atlas and Tabula Sapiens have made multi-organ reference maps a routine part of biomedical research [1–5]. These atlases are valuable only when transcriptional profiles can be assigned to biologically meaningful cell identities. Cell-type annotation is therefore not a cosmetic post-processing step: it determines the units used for differential abundance, state-specific expression, trajectory analysis, cell–cell communication, and association with clinical phenotypes. Manual marker-based annotation remains widely used, but it is labour intensive, sensitive to the available expertise and marker list, and difficult to reproduce at atlas scale. General best-practice and reporting frameworks consequently treat annotation provenance and reproducible preprocessing as essential parts of a single-cell analysis [6–8].

Automated annotation methods have approached this problem through several statistical traditions. SingleR and scmap project cells onto labelled references; scPred uses a supervised support-vector framework; CellTypist uses regularized logistic regression; Azimuth combines reference mapping with harmonized single-cell resources; and latent-variable models such as scVI learn lower-dimensional representations that can support downstream annotation [9–14]. Cell BLAST, scNym, Harmony and scArches further illustrate the range from learned search spaces and semi-supervised classification to integration and transfer learning [15–18]. Broad benchmarks have repeatedly shown that relatively simple supervised models can be difficult to outperform when training and test data share protocols and labels [19, 20]. Their success does not remove the need for transferable representations, because curated labels can be scarce, datasets can differ by donor or platform, and cell states may not align cleanly between references. It does, however, establish that new representation-learning methods require carefully matched baselines.

Single-cell foundation models extend this representation-learning idea by pretraining transformer-like architectures on millions of transcriptomes. scBERT, Geneformer, scGPT, scFoundation, Universal Cell Embeddings and CellFM use different tokenization schemes and pretraining objectives, but share the aim of learning reusable cell or gene representations [21–27]. scGPT is a prominent example. Its whole-human checkpoint was trained on a large CELLxGENE-derived corpus and exposes an interface for obtaining fixed-dimensional cell embeddings that can be used by a separate task-specific learner [23, 28]. Pretraining could be most useful when target labels are limited, when biological variation must be transferred between studies, or when the downstream task benefits from structure learned outside the target cohort. Yet pretraining alone does not produce an annotation probability. A frozen encoder must be coupled to a classifier, and the resulting performance and calibration belong to that complete pipeline.

This distinction is central to fair evaluation. If scGPT embeddings are paired only with a linear classifier while expression features are paired with a nonlinear learner, the comparison confounds representation and classifier capacity. Conversely, a classical model given a larger or differently selected gene set may benefit from an information advantage. Feature selection performed before a train–test split can leak information from the test partition even when class labels are not used. Sparse matrices create a further implementation hazard for some tree libraries because omitted biological zeros can be interpreted as missing values. A controlled benchmark should therefore match the genes available to both representations, apply the same classifier families, learn all filters within the training partition, preserve biological zeros, and name each representation–classifier combination explicitly. These safeguards also align with broader calls for transparent reporting of single-cell experiments and computational transformations [8].

Generalization also depends on the unit of splitting. Cells sampled from one donor share genetics, exposures, processing, and technical effects; assigning cells from the same donor to both training and test sets can yield an optimistic estimate of performance on unseen individuals. The problem is not unique to scRNA-seq, but is amplified by the large ratio of cells to biological replicates. Cell-level bootstrap intervals likewise treat correlated cells as independent and can appear much narrower than uncertainty obtained by resampling donors. Benchmarks of single-cell integration and foundation-model embeddings have shown that batch and donor structure can dominate apparent biological organization [29, 30]. More generally, treating cells as independent replicates can produce misleading precision when biological replication occurs at sample or donor level [31]. Donor-held-out evaluation is therefore a necessary complement to large within-atlas cell-level comparisons, although it still does not substitute for external-atlas transfer when protocols and annotation systems differ.

Probability quality is a second, distinct requirement. Accuracy and Macro F1 describe class discrimination, but many annotation workflows also use confidence scores to reject uncertain cells, prioritize manual review, or weight downstream analyses. Calibration asks whether predicted probabilities correspond to empirical frequencies. Expected calibration error (ECE) is intuitive but depends on bin definitions and is not a proper scoring rule [32–34]. Adaptive and classwise ECE expose different aspects of multiclass calibration, whereas the Brier score and negative log-likelihood evaluate the full probability vector and reward honest probabilistic prediction [35, 36]. Reliability diagrams are most interpretable when bin counts and uncertainty are shown. Post-hoc approaches range from a single temperature to multiclass calibrators, but require calibration observations that remain separate from evaluation data [37, 38]. Temperature scaling provides a simple test of whether probability sharpness, rather than class ranking, accounts for observed miscalibration [32].

Independent evaluations have begun to question whether large single-cell models consistently improve over simpler representations. Zero-shot studies found that scGPT and Geneformer did not uniformly outperform highly variable genes, Harmony, or scVI for cell-type structure and batch integration, and that results depended strongly on dataset and task [30]. Other benchmarking frameworks have emphasized inconsistent software interfaces, hyperparameter sensitivity, and the need to distinguish zero-shot, frozen-probe, and fine-tuned settings [39–41]. These observations do not imply that pretraining lacks value. They instead reinforce a general principle: the benefit of a foundation model must be demonstrated for a specified representation, downstream learner, label regime, and biological split.

Our original submission addressed accuracy and calibration across six large atlases but used a single cell-level split, test-informed feature selection, a custom scGPT wrapper, and an incomplete calibration assessment. A full executable audit during revision showed that the wrapper did not reproduce the released scGPT embedding recipe and that sparse XGBoost inputs could treat zero counts as missing. We therefore superseded the earlier values and reran the benchmark from raw counts using the official scGPT 0.2.5 embedding helper, training-only Seurat v3 feature selection, exact vocabulary matching, dense XGBoost inputs, and a factorial comparison of two representations by three classifiers. This correction narrows the claims but makes the numerical record reproducible.

The revised study addresses four questions. First, how do official frozen scGPT embeddings compare with normalized expression of the same genes when both are paired with Logistic Regression, Random Forest, and XGBoost? Second, how do conclusions change when donors rather than cells define five independent train–test partitions? Third, does reducing the number of labelled training cells reveal an advantage for the pretrained representation? Fourth, how stable are calibration conclusions across complementary metrics and after temperature scaling fitted without access to test donors? We answer these questions across six public atlases containing 1,405,933 cells, with a focused donor-held-out analysis in 199 Indonesian donors. The results support strong classical baselines for the tested supervised settings while defining, rather than dismissing, the transfer regimes that remain open.

2 Results

2.1 A corrected factorial benchmark replaced the original comparison

The six atlases contained 1,405,933 cells; every source label exceeded the prespecified five-cell threshold (Table 1). For each atlas, we generated one stratified 80/20 cell-level partition with random seed 42. All gene filtering and Seurat v3 highly variable gene selection were then performed using the training partition only. Genes were matched to the released scGPT vocabulary, and the same final gene set was used to generate the official frozen scGPT embeddings and the normalized log-expression representation. Twenty-one TNBC cells, three brain cells, and 1,381 multi-tissue TME cells had zero total input across the shared gene set and were excluded identically from both representations; 1,404,528 cells remained, including 280,912 test cells.

The controlled design crossed two representations with three downstream classifiers. The scGPT branch used 512-dimensional cell embeddings returned by the official scGPT 0.2.5 helper. The matched branch used library-size-normalized and log-transformed expression of the exact same training-selected, vocabulary-compatible genes. Logistic Regression, Random Forest, and XGBoost were fit with the same settings on both representations. XGBoost received dense arrays so that a zero count remained a numerical zero rather than a sparse missing value. This design produced 36 primary pipelines and replaced the original custom-wrapper, unmatched-feature, and fine-tuning results.

2.2 Matched expression produced the highest observed Macro F1 in all six atlases

Figure 1 presents the complete accuracy and ECE matrices. The strongest matched-expression pipeline achieved higher observed Macro F1 than the strongest frozen-scGPT pipeline in each atlas (Table 2). The absolute difference was smallest in TNBC, where Matched expression + XGBoost reached 0.992 compared with 0.980 for scGPT embeddings + XGBoost. Differences were 0.020 in Indonesia PBMC, 0.021 in the brain atlas, and 0.051 in the multi-tissue TME atlas. They were larger in pancreas: 0.983 versus 0.874 in human and 0.993 versus 0.865 in pig. Thus, the earlier numerical impression that multi-tissue TME was effectively tied did not survive the corrected official pipeline; matched-expression XGBoost reached 0.416 compared with 0.365 for scGPT embeddings + XGBoost.

The cell-bootstrap intervals quantify uncertainty conditional on each fitted training partition. On TNBC and Indonesia PBMC, the intervals around the strongest pipelines were narrow because the test sets contained 85,561 and 92,407 cells. Brain and pancreas intervals were wider because their class-specific test counts were smaller. The paired cell-bootstrap file supplied with the revision retains the same test resample for the two representations, avoiding the loss of information that occurs when two marginal intervals are compared. Because these resamples treat cells as exchangeable, they are not interpreted as population-level biological uncertainty; donor-resampling results are reported separately below.

Classifier choice materially changed performance within each representation. XGBoost was the strongest frozen-scGPT head in all six atlases, demonstrating that the discriminative information in the frozen embeddings was not adequately characterized by a linear probe alone. The best matched classifier depended on the dataset: XGBoost was strongest for TNBC, brain, and multi-tissue TME, whereas Logistic Regression was strongest for Indonesia PBMC and both pancreas atlases. Random Forest was rarely strongest and was particularly weak for matched expression in the brain atlas. These results rule out language that treats either "scGPT" or "classical models" as a single estimator. Every comparison below refers to a specified representation–classifier pipeline.

2.3 Calibration depended on the complete representation and classifier pipeline

The calibration results did not reduce to one universal ordering. Among the strongest pipelines displayed in Table 2, scGPT embeddings + XGBoost had lower equal-width ECE in TNBC, Indonesia PBMC, and multi-tissue TME, whereas the strongest matched pipeline had lower ECE in brain, human pancreas, and pig pancreas. In contrast, the matched pipeline had lower Brier score and NLL in all six datasets. For example, in Indonesia PBMC the scGPT XGBoost ECE was 0.004 versus 0.009 for matched Logistic Regression, but its Brier score was 0.112 versus 0.072 and its NLL was 0.200 versus 0.133. In multi-tissue TME, ECE favoured scGPT XGBoost (0.010 versus 0.023), while both proper scoring rules favoured matched XGBoost. A statement that one representation was simply "better calibrated" would therefore conceal metric-specific

Table 1: Public atlases and analysis partitions in the corrected benchmark. Source cells are counted before shared-input quality control.

Dataset	Source	Excluded	Retained	Train	Test	Classes	Genes
TNBC	427,823	21	427,802	342,241	85,561	7	1,165
Indonesia PBMC	462,034	0	462,034	369,627	92,407	13	1,135
Brain atlas	75,583	3	75,580	60,464	15,116	10	939
Multi-tissue TME	391,963	1,381	390,582	312,461	78,121	11	1,067
Human pancreas	26,474	0	26,474	21,179	5,295	4	1,149
Pig pancreas	22,056	0	22,056	17,644	4,412	4	1,104
Total	1,405,933	1,405	1,404,528	1,123,616	280,912	–	–

No cell was removed by the prespecified rare-label filter. Excluded cells had zero total counts across the training-selected genes that could be matched to the scGPT vocabulary. "Genes" is the final vocabulary-matched feature count. The displayed train and test counts refer to the primary stratified 80/20 within-atlas split; the donor-held-out experiment used separate fold-specific partitions.

Table 2: Strongest frozen-scGPT and matched-expression pipelines in each within-atlas benchmark. Selection is based on observed Macro F1 within each representation.

Dataset	Representation	Classifier	Macro F1	95% CI	ECE	Brier	NLL
TNBC	scGPT embeddings	XGB	0.980	0.979–0.981	0.001	0.036	0.069
	Matched expression	XGB	**0.992**	0.991–0.992	0.002	**0.015**	**0.030**
Indonesia PBMC	scGPT embeddings	XGB	0.936	0.933–0.938	**0.004**	0.112	0.200
	Matched expression	LR	**0.956**	0.953–0.958	0.009	**0.072**	**0.133**
Brain atlas	scGPT embeddings	XGB	0.931	0.904–0.953	0.008	0.021	0.060
	Matched expression	XGB	**0.952**	0.929–0.970	**0.005**	**0.019**	**0.043**
Multi-tissue TME	scGPT embeddings	XGB	0.365	0.359–0.372	**0.010**	0.644	1.442
	Matched expression	XGB	**0.416**	0.404–0.430	0.023	**0.614**	**1.368**
Human pancreas	scGPT embeddings	XGB	0.874	0.817–0.920	0.005	0.016	0.035
	Matched expression	LR	**0.983**	0.962–0.999	**0.001**	**0.003**	**0.005**
Pig pancreas	scGPT embeddings	XGB	0.865	0.810–0.907	0.011	0.043	0.087
	Matched expression	LR	**0.993**	0.989–0.997	**0.002**	**0.004**	**0.011**

CI, 5,000-draw cell bootstrap confidence interval; ECE, equal-width 10-bin expected calibration error; Brier, multiclass Brier score; NLL, negative log-likelihood; LR, Logistic Regression; XGB, XGBoost. Bold indicates the better value within a dataset and metric among the two displayed pipelines. Complete results for all 36 pipelines are reported in Supplementary Table S1.

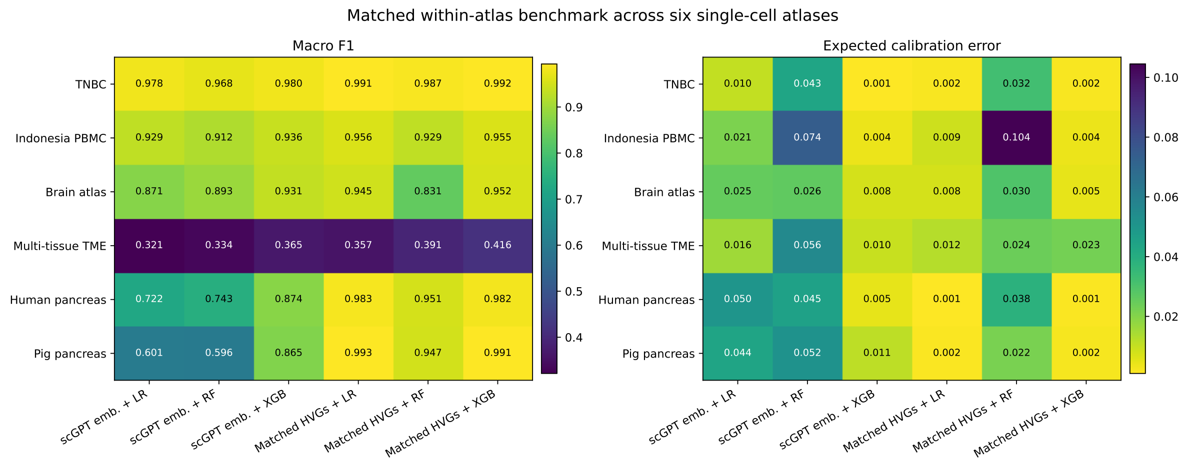

Figure 1: Controlled within-atlas benchmark across six datasets. Heatmaps report Macro F1 and equal-width 10-bin ECE for official frozen scGPT embeddings and matched normalized expression features, each coupled to Logistic Regression, Random Forest, or XGBoost. All feature selection was learned from the training partition.

behavior.

The downstream learner also changed probability quality while the representation remained fixed. Random Forest often had substantially higher ECE than Logistic Regression or XGBoost, including 0.104 for matched expression in within-atlas Indonesia PBMC and 0.030 for matched expression in brain. XGBoost generally produced low confidence ECE for frozen scGPT embeddings, even when discrimination remained below the matched alternative. This confirms the reviewers' central point: predicted probabilities are generated by the downstream head and cannot be interpreted as intrinsic probabilities of the pretrained encoder.

Complete equal-width, adaptive, and classwise ECE values, Brier scores, NLL values, and cell-bootstrap intervals are reported in Supplementary Table S1. Reliability plots for all atlases show the number of cells in each confidence bin and Wilson 95% intervals for observed accuracy (Supplementary Figures S1–S6). Empty bins are retained with count zero rather than silently omitted. Together, these outputs show where a small aggregate ECE is supported by large populated bins and where estimates depend on a limited range of confidences.

2.4 Five donor-held-out folds tested generalization to unseen individuals

The Indonesia PBMC metadata contained a one-to-one mapping between `donor_id` and `sample_id` for 199 donors. We used five-fold stratified group cross-validation, assigning every cell from a donor wholly to one outer fold. Test folds contained 38–42 donors and 90,015–93,930 cells; no donor appeared in both training and test data. Gene filtering, highly variable gene selection, vocabulary matching, both representations, and every classifier were recomputed independently in every fold.

The strongest frozen-scGPT pipeline was scGPT embeddings + XGBoost, with pooled out-of-fold Macro F1 of 0.934 and fold mean 0.934 ± 0.005 (Figure 2; Table 3). The strongest matched pipeline by Macro F1 was Matched expression + Logistic Regression, with pooled Macro F1 of 0.956 and fold mean 0.956 ± 0.003. Its observed paired difference from scGPT embeddings + XGBoost was 0.02236. A 5,000-draw paired donor bootstrap gave a 95% interval of 0.02066–0.02423, and the difference was positive in every bootstrap draw. When representation comparisons were matched by classifier, matched expression also exceeded scGPT embeddings for Logistic Regression, Random Forest, and XGBoost, with paired differences of 0.02870, 0.01279, and 0.02099, respectively.

The relative results were stable across outer folds. Matched expression + Logistic Regression ranged from 0.954 to 0.960 Macro F1, while scGPT embeddings + XGBoost ranged from 0.929 to 0.940. Matched expression + XGBoost was close to matched Logistic Regression and produced the lowest NLL (0.125) and Brier score (0.071) in pooled predictions. The distinction matters: Logistic Regression had the highest pooled Macro F1, whereas XGBoost produced the best proper probability scores. The donor-held-out analysis therefore reproduces the matched-expression advantage without requiring a claim that one classifier dominates every metric.

Table 3: Five-fold donor-held-out evaluation on Indonesia PBMC.

Pipeline	Pooled F1	Fold F1	Donor CI	Cell CI	ECE	Brier	NLL
scGPT embeddings + LR	0.928	0.928 ± 0.003	0.924–0.931	0.926–0.929	0.022	0.114	0.210
scGPT embeddings + RF	0.912	0.912 ± 0.005	0.907–0.916	0.910–0.913	0.072	0.154	0.295
scGPT embeddings + XGB	0.934	0.934 ± 0.005	0.931–0.937	0.933–0.935	0.003	0.112	0.201
Matched expression + LR	**0.956**	$\mathbf{0.956 \pm 0.003}$	0.954–0.958	0.955–0.958	0.009	0.072	0.133
Matched expression + RF	0.925	0.925 ± 0.003	0.921–0.928	0.923–0.927	0.106	0.110	0.231
Matched expression + XGB	0.955	0.955 ± 0.003	0.953–0.958	0.954–0.956	0.004	**0.071**	**0.125**

Fold F1 is mean ± standard deviation across five outer donor folds. Donor and cell CIs are 5,000-draw bootstrap intervals for pooled Macro F1. Donor resampling carries all cells from a selected donor together. LR, Logistic Regression; RF, Random Forest; XGB, XGBoost.

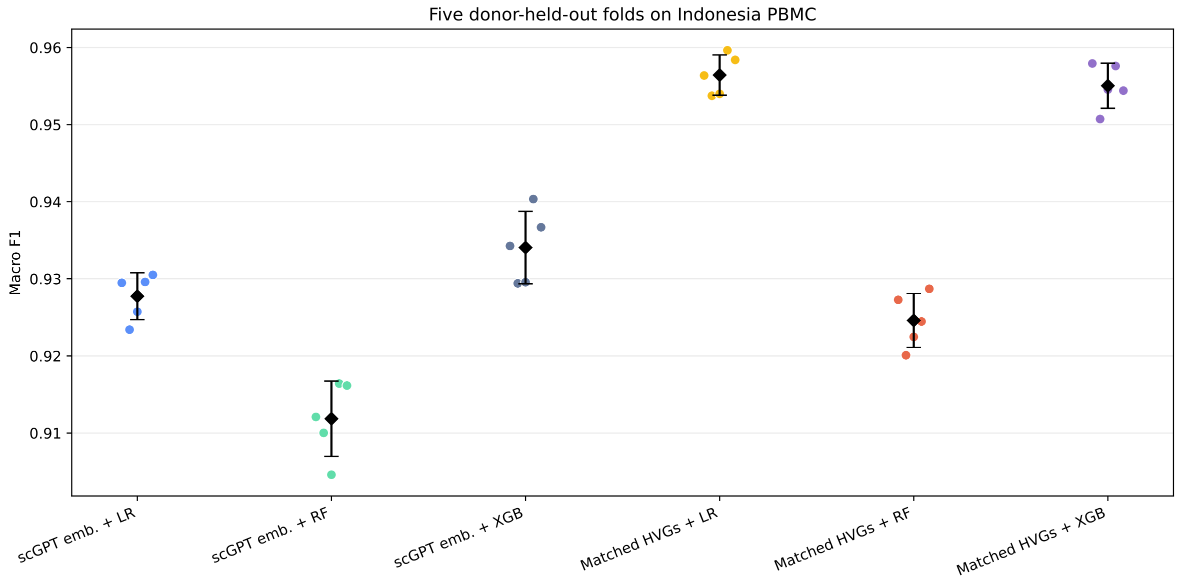

Figure 2: Five donor-held-out folds on Indonesia PBMC. Points are fold-specific Macro F1 values; diamonds and error bars show mean ± standard deviation across folds. All 199 donors contribute to exactly one test fold.

2.5 Donor resampling produced wider uncertainty than cell resampling

For every donor-held-out pipeline, the donor-cluster Macro F1 interval was wider than the interval obtained by independently resampling cells. For Matched expression + Logistic Regression, the donor interval was 0.954–0.958 compared with 0.955–0.958 under cell resampling. For scGPT embeddings + Random Forest, the corresponding intervals were 0.907–0.916 and 0.910–0.913. Across all six pipelines, donor-interval widths were approximately 1.7–3.1 times the cell-bootstrap widths. The point estimates and model ordering remained stable, but treating cells as independent overstated precision.

Donor-level reliability plots used the same ten confidence bins as the within-atlas plots but calculated accuracy uncertainty by resampling donors and carrying all their cells together (Supplementary Figure S7). This analysis directly connects dependence in the data to uncertainty in calibration curves. It also clarifies the scope of the within-atlas intervals: cell bootstraps describe variation among observed test cells for a fixed training set, whereas donor bootstraps better approximate variation across biological sampling units in the cohort for which donor identifiers were available.

2.6 Matched expression remained stronger throughout the tested label-scarcity range

To test a setting in which pretraining might be expected to help, we reduced the number of labelled training cells within each outer donor fold to 10, 25, 50, 100, 250, or 500 cells per class. Sampling was performed without replacement using five seeds, and test donors were unchanged. Logistic Regression and XGBoost were evaluated on both representations, producing 600 fitted scarcity experiments.

Performance increased with the label budget for all pipelines (Figure 3; Table 4). With Logistic Regression and ten labelled cells per class, mean Macro F1 was 0.679 for scGPT embeddings and 0.856 for matched expression, a difference of 0.177. At 100 labels per class, the values were 0.817 and 0.913; at 500, they were 0.863 and 0.915. XGBoost showed smaller but consistently positive matched-expression differences, from 0.026 at ten labels per class to 0.042 at 500. No tested classifier, label budget, outer fold, and aggregated setting established a crossover in favour of frozen scGPT embeddings.

Calibration under scarcity depended strongly on the classifier. Logistic Regression on scGPT embeddings was under-confident at small sample sizes, with mean ECE 0.537 at ten labels per class, and improved gradually as the budget increased. XGBoost produced lower ECE for both representations at the smallest budgets but lower Macro F1 than matched Logistic Regression. These results concern supervised adaptation within one PBMC cohort. They do not evaluate zero-shot use, previously unseen label categories, external-atlas transfer, or extremely rare populations whose labels cannot support stratified sampling.

2.7 Temperature scaling improved probability quality without changing predictions

Temperature scaling was evaluated for the two Logistic Regression pipelines. For each outer fold, one scalar temperature was optimized on logits from an inner group split containing training donors only and was then applied once to the untouched outer-test donors. The mean fitted temperature was 0.806 for scGPT embeddings, indicating that the unscaled probabilities were too diffuse, and 1.233 for matched expression, indicating modest over-confidence.

For scGPT embeddings + Logistic Regression, mean ECE decreased from 0.0240 to 0.0030 and mean NLL from 0.2151 to 0.2065 (Figure 4; Table 4). The Brier score decreased from 0.1165 to 0.1152. For matched expression + Logistic Regression, ECE decreased from 0.0102 to 0.0022, NLL from 0.1379 to 0.1340, and Brier score from 0.0730 to 0.0725. Macro F1 was unchanged for both pipelines because scalar temperature scaling preserves the ordering of class logits.

These results show that a substantial component of the observed ECE can be corrected without modifying the encoder, expression representation, classifier coefficients, or predicted classes. They do not erase the discrimination difference between representations. Rather, they demonstrate why annotation studies should report the representation, classifier, fitting protocol, and calibration procedure together.

3 Discussion

This revision provides a controlled assessment of accuracy and probability quality in annotation pipelines built from frozen scGPT embeddings. Across six atlases and three downstream classifiers, normalized expression of the same training-selected genes supplied to scGPT produced the strongest observed Macro F1 in every dataset. The conclusion is narrower than a statement that classical machine learning is universally superior to foundation models. It applies to supervised within-atlas annotation with the released whole-human checkpoint, broad curated labels, and the specified classifiers. Within those conditions, however, the comparison is stringent: neither a larger classical feature set nor a weaker scGPT head explains the result.

The classifier-matched design changes how the result should be interpreted. XGBoost substantially improved frozen-scGPT performance relative to Logistic Regression and was the strongest scGPT head in every atlas. TNBC scGPT embeddings + XGBoost reached Macro F1 0.980, and the same pipeline reached 0.936 in within-atlas Indonesia PBMC and 0.934 in donor-held-out PBMC. These are strong annotation results, not evidence that the embeddings lack cell-type information. The remaining question is whether the compressed representation adds predictive information beyond the matched expression measurements from which it is derived. Under the tested conditions it did not: access to normalized expression retained or exposed information that the downstream learners used more effectively.

Several mechanisms could contribute. The scGPT cell embedding compresses as many as 1,200 gene-level measurements into 512 dimensions under a pretraining objective that is not optimized specifically for the target cell labels. Compression may discard variation that is highly predictive

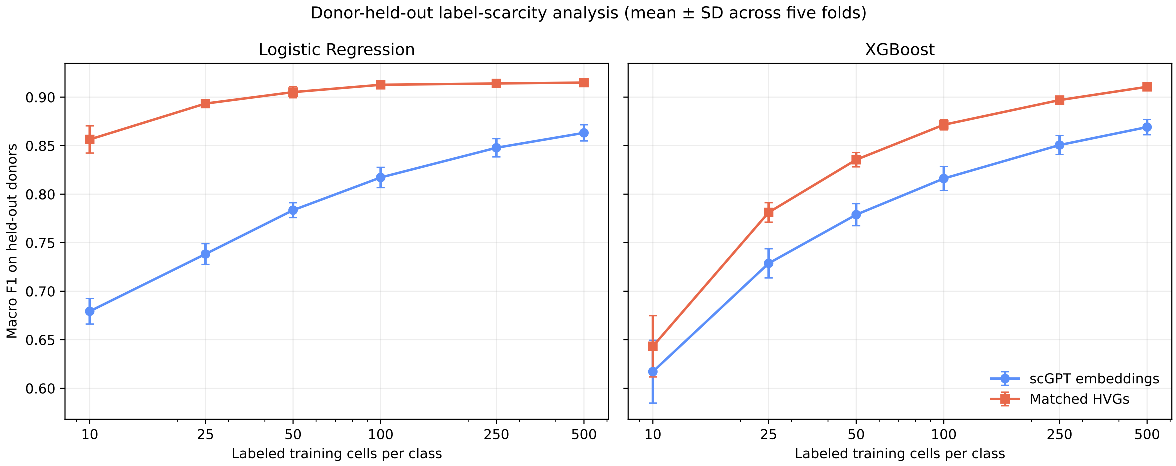

Figure 3: Donor-held-out label-scarcity analysis. Curves show mean Macro F1 $\pm$ standard deviation across five outer donor folds after averaging five within-fold sampling seeds at each label budget. Test donors remain fixed within each outer fold.

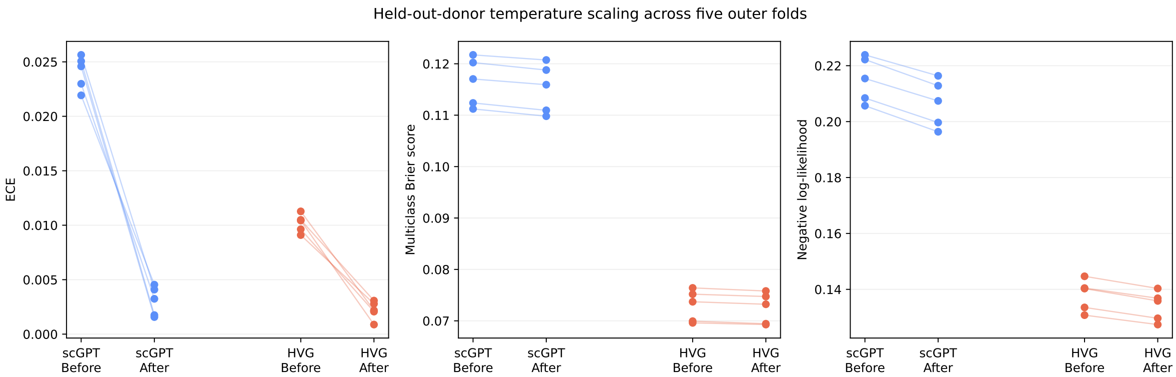

Figure 4: Held-out-donor temperature scaling. Paired points show calibration metrics before and after scaling. A scalar temperature was fitted using only outer-training donors and applied to the corresponding outer-test donors.

Table 4: Label scarcity and held-out-donor temperature scaling on Indonesia PBMC.

A. Logistic Regression label-scarcity analysis				
Labels per class	scGPT F1	Matched F1	Difference	scGPT / matched ECE
10	0.679 ± 0.013	0.856 ± 0.014	0.177	0.537 / 0.065
25	0.738 ± 0.011	0.893 ± 0.002	0.155	0.532 / 0.014
50	0.784 ± 0.008	0.905 ± 0.006	0.122	0.494 / 0.016
100	0.817 ± 0.010	0.913 ± 0.003	0.096	0.430 / 0.029
250	0.848 ± 0.009	0.914 ± 0.002	0.066	0.323 / 0.043
500	0.863 ± 0.008	0.915 ± 0.003	0.052	0.246 / 0.050
B. Temperature scaling of donor-held-out Logistic Regression				
Pipeline and state	Macro F1	ECE	Brier	NLL
scGPT embeddings before	0.925 ± 0.003	0.024	0.117	0.215
scGPT embeddings after	0.925 ± 0.003	**0.003**	0.115	**0.207**
Matched expression before	0.956 ± 0.003	0.010	0.073	0.138
Matched expression after	0.956 ± 0.003	**0.002**	0.072	**0.134**

Scarcity values are fold means $\pm$ standard deviations after averaging five sampling seeds within each of five outer donor folds; each budget was capped by available cells without replacement. Temperature scaling used an inner donor split from outer-training donors only. Complete scarcity results for Logistic Regression and XGBoost are in Supplementary Table S3.

for a particular atlas while preserving more general structure. Broad target labels can also be resolved by canonical marker combinations for which direct expression is already sufficient. In this regime, a supervised classifier with hundreds of thousands of labelled cells has little need for an external prior. These explanations are consistent with independent zero-shot evaluations in which highly variable genes, scVI, or Harmony often matched or exceeded foundation-model embeddings [30]. They should be treated as hypotheses about task alignment rather than evidence against pretraining in general.

The donor-held-out experiment strengthens the practical relevance of the comparison. Its five folds exclude direct overlap of donors and repeat all feature-selection and fitting steps. Matched expression remained ahead, with a paired donor-bootstrap difference of approximately 0.022 Macro F1 between the best pipelines. At the same time, holding out donors from one cohort is not equivalent to transfer across studies. Library preparation, processing, cell-type ontology, ancestry composition, and laboratory effects remain shared between outer folds. A truly external benchmark would train on one atlas and apply a harmonized label system to an independently generated atlas. We therefore describe the experiment as generalization to unseen donors, not as out-of-cohort or zero-shot generalization.

The comparison between donor and cell bootstraps demonstrates why the biological unit matters even when point estimates are based on hundreds of thousands of cells. Donor-cluster intervals were wider for all six pipelines because correlated cells moved together between resamples. This is consistent with evidence that treating cells instead of biological samples as replicates can inflate apparent certainty in single-cell inference [31]. The ranking remained stable in this cohort, but that outcome could not have been assumed from the cell-level analysis. Cell bootstrap intervals remain useful as conditional descriptions of a fixed fitted model and observed cell mixture; they should not be presented as if the number of cells were the number of independent biological replicates. Similar considerations apply to differential abundance and other single-cell analyses in which inference can be dominated by pseudoreplication.

The scarcity experiment did not reveal the expected advantage of pretraining. Matched expression was stronger at every tested budget from 10 to 500 labelled cells per class, and the largest Logistic Regression difference occurred at the smallest budget. One interpretation is that the selected broad PBMC classes require relatively simple marker information and that even ten examples per class provide enough supervision for a regularized linear model. Another is that frozen scGPT embeddings are not aligned with this donor-held-out label system. Importantly, this experiment subsamples labelled cells from a cohort for which the set of target classes is already known. It does not test a new cell type, an ontology mismatch, a zero-label task, a single rare population embedded in a much larger negative class, or transfer between laboratories. Those remain settings in which pretraining may offer greater value.

Calibration is governed by the complete prediction pipeline. The lowest equal-width ECE sometimes belonged to scGPT embeddings + XGBoost and sometimes to a matched-expression pipeline, whereas Brier score and NLL favoured matched expression for the strongest pipelines in all six atlases. ECE summarizes gaps after binning only the winning-class confidence, so a low value can coexist with weaker discrimination or less informative full probability vectors [33, 34].

Proper scoring rules capture different information and should accompany ECE. Random Forest illustrates the dependence on the head particularly clearly: it often produced higher ECE than Logistic Regression or XGBoost on the same representation. Claims about "scGPT calibration" without specifying the downstream classifier are therefore under-determined.

Temperature scaling provides a constructive result. A single parameter fitted without test-donor access reduced ECE and NLL for both Logistic Regression pipelines while leaving Macro F1 unchanged. Thus, part of the apparent miscalibration was a correctable mismatch in probability sharpness rather than an error in class ordering. Post-hoc calibration does not improve representation quality, recover missed cell populations, or remove domain shift, and it requires a representative calibration set. Nevertheless, it is inexpensive and appropriate when probabilities will be used for rejection, triage, or weighting. Reliability plots with bin counts and uncertainty should be retained because a scalar score cannot show whether a model fails in densely or sparsely populated confidence regions.

The computational audit is itself an important result. The original custom wrapper used preprocessing and value-binning choices that did not reproduce the released embedding interface, and sparse XGBoost inputs risked treating biological zeros as missing. Replacing rather than retrofitting those values materially changed the numerical record, including the multi-tissue TME comparison. Foundation-model benchmarks combine data preprocessing, tokenization, checkpoint configuration, representation extraction, and downstream fitting; a plausible description of one component is not sufficient for reproducibility. Source commits, checkpoint hashes, intermediate audits, and executable validation should be considered part of the method.

Several limitations remain. Only scGPT 0.2.5 and the released whole-human checkpoint were evaluated; results need not transfer to Geneformer, scFoundation, UCE, newer checkpoints, or models with different pretraining objectives. We evaluated frozen embeddings rather than end-to-end fine-tuning under the corrected official recipe. The six primary atlas analyses use one within-atlas training partition each, and their cell-bootstrap intervals do not include training-set variability. Donor metadata supported a stronger analysis in only one cohort. The target labels are curated and broad, ranging from four to thirteen classes; every source label exceeded the prespecified five-cell threshold, so the benchmark does not test the behavior of that exclusion rule. Hyperparameter exploration was deliberately limited, so results characterize specified practical pipelines rather than the best possible configuration of each algorithm.

We also did not test external-atlas transfer, ontology harmonization, zero-shot mapping, extremely rare cell types, fine-grained states, perturbation response, multimodal integration, or gene-network inference. Some benchmark datasets may overlap partly with the large public corpus used for scGPT pretraining, and public checkpoint documentation does not permit every individual pretraining cell to be traced. Calibration estimates depend on binning, while Brier score and NLL combine calibration and discrimination. Finally, donor bootstrap reflects the empirical donor distribution of the Indonesia cohort and cannot represent biological populations absent from it.

4 Conclusions

Strong classical baselines remain essential when evaluating single-cell foundation models. In the tested supervised within-atlas and donor-held-out settings, matched normalized expression paired with standard classifiers provided the best annotation accuracy, while probability quality depended on the downstream head and calibration procedure. The corrected results do not establish that pretrained representations lack value; they identify conditions in which their additional complexity did not improve over direct expression features. Future claims of foundation-model advantage should be evaluated under biological holdout, label scarcity, matched downstream learners, proper scoring rules, and genuinely external transfer designs.

5 Methods

5.1 Study design

We designed the revision as a factorial comparison between two input representations and three supervised classifiers. The first representation was the frozen cell embedding returned by the official scGPT embedding interface. The second was normalized expression of exactly the genes supplied to that interface, hereafter "matched expression". Each representation was paired with Logistic Regression, Random Forest, and XGBoost. Consequently, classifier effects could be distinguished from representation effects without giving the matched-expression pipelines a larger gene set. The primary endpoint was Macro F1. Accuracy and balanced accuracy were secondary discrimination measures; equal-width, adaptive, and classwise expected calibration error (ECE), multiclass Brier score, and negative log-likelihood (NLL) characterized probability quality. The analysis plan was expanded in response to peer review to include donor-held-out validation, donor-cluster uncertainty, label scarcity, and post-hoc temperature scaling.

The study used only public, de-identified data and no outcome was used to tune a representation or classifier hyperparameter. Hyperparameters were fixed before the complete benchmark was aggregated. Within-atlas analyses used one prespecified split to retain comparability across all six datasets. The Indonesia PBMC cohort was then used for five biologically grouped outer folds, providing a separate assessment of generalization to unseen donors and variability between partitions.

5.2 Single-cell datasets and target labels

Six curated AnnData objects were obtained from CZ CELL×GENE Discover [28]: untreated triple-negative breast cancer (TNBC), Indonesia peripheral blood mononuclear cells (PBMC) [42], an amyotrophic lateral sclerosis brain atlas [43], a multi-tissue tumour microenvironment (TME) atlas [44], human pancreatic islets, and pig pancreatic islets [45]. Stable collection and dataset identifiers are listed in the Data availability statement. Together, the source matrices contained 1,405,933 cells. The benchmark used the portal-provided `cell_type` annotation as the target and raw integer counts from `adata.raw`. Labels represented by five or fewer cells would have been removed before partitioning because they could not support the stratified primary split; no source label met this exclusion criterion. A total of 1,405 cells had no non-zero count in the final shared input space, leaving 1,404,528 cells (Table 1).

The six datasets span immune, neural, tumour, stromal, and endocrine compartments and contain 4–13 target classes. They were chosen to reproduce the biological breadth of the original submission while supporting a uniform, auditable input representation. The labels are nevertheless mostly broad cell categories; the benchmark is not a test of fine cell states or harmonization between independently defined ontologies. The Indonesia object contained 462,034 cells, 13 classes, and 199 non-missing values of `donor_id`; `donor_id` and `sample_id` had a one-to-one mapping. Donor identity, rather than sequencing batch, was therefore used as the grouping unit.

5.3 Within-atlas partitioning and feature selection

For each atlas, cells were divided into 80% training and 20% test partitions by stratified random sampling with seed 42. All subsequent gene filtering and feature selection were restricted to the training partition. Starting from raw counts, genes with fewer than three total training counts were excluded, and the 1,200 highest-ranked variable genes were selected with the Seurat v3 variance-stabilizing procedure as implemented in Scanpy [46, 47]. Gene symbols were compared with the vocabulary of the released whole-human scGPT checkpoint. Exact matches were used first; an upper-case symbol was used when the original symbol was absent and the upper-case symbol was present. Duplicates introduced by mapping were removed by retaining the higher-ranked gene. This produced 939–1,165 genes per atlas.

The same ordered set of vocabulary-matched genes was used for both representations. A cell with zero total count across this set could not yield a meaningful scGPT input and was excluded from both branches before either model was fitted. The zero-input filter was therefore representation-neutral. The final train and test counts in Table 1 reflect this shared filter. No test label, test count statistic, or test-derived highly variable gene rank entered training feature selection.

5.4 Official frozen scGPT representation

Frozen embeddings were generated with scGPT 0.2.5 from source commit `cebd6fa`, using the authors' public whole-human checkpoint and the official `scgpt.tasks.embed_data` function [23]. The input was the raw-count matrix restricted to the fold-specific matched genes, with gene symbols supplied in `feature_name`. We used `max_length=1200`, batch size 256, the fast transformer path, and an NVIDIA H100 80 GB GPU. The helper performed the checkpoint-defined value preprocessing and returned one 512-dimensional, unit-normalized cell vector. The encoder weights were never updated, and each downstream classifier was trained only on the training cell vectors.

Before the full run, a 512-cell GPU smoke test verified the package version, CUDA execution, output dimensions, finite values, and unit norms. The three checkpoint files were recorded by SHA-256: `args.json`, `c18e075e018140cb8b2d9029387b9de26607a5ce6a8ccabd6ead70cd76b95d60`;

`vocab.json`, `acca93d114ca62c3f0f50debbd23e8c87f0714f4737764454f6b2b13f2e8580f`; and `best_model.pt`, `6cb5d451ab5c4b33eb673adbe4fddc61d2389df1b89b7651a9fe2e557572b922`. These identifiers distinguish the evaluated checkpoint and extraction path from custom wrappers or later releases.

5.5 Matched-expression representation

For the matched branch, the same raw counts and gene order were library-size normalized to 10,000 counts per cell and transformed with $\log(1 + x)$. This common preprocessing preserves expression zeros and is widely used for supervised single-cell analysis [13, 47]. The resulting matrix contained no gene unavailable to scGPT and no gene selected from the test set. Thus, comparisons estimate the effect of replacing direct matched expression with the frozen embedding, conditional on the same initial measurements and downstream learner.

5.6 Downstream classifiers

We fitted three multi-class classifiers without dataset-specific tuning. Logistic Regression used L_2 regularization with $C = 1$, the `lbfgs` solver, tolerance 10^{-4}, and at most 3,000 iterations. A convergence warning was treated as an error. Random Forest used 200 trees, Gini splits, square-root feature sampling, and bootstrap sampling. XGBoost used 200 boosting rounds, maximum depth 6, learning rate 0.1, the multi-class soft-probability objective, histogram tree construction, and CUDA acceleration [48, 49]. Seeds were fixed in the executable analysis. All models returned a probability vector over the same alphabetically encoded class set.

Sparse matrices require special care in XGBoost because an absent sparse entry can be interpreted as missing rather than as a measured zero. Both XGBoost training and test matrices were therefore materialized as dense 32-bit arrays, ensuring identical zero semantics for scGPT and matched-expression inputs. Logistic Regression and Random Forest retained numerically equivalent sparse or dense views as supported by their implementations. Scikit-learn 1.9.0 and XGBoost 3.2.0 were used [49, 50].

5.7 Five-fold donor-held-out evaluation

We evaluated all six pipelines on Indonesia PBMC with `StratifiedGroupKFold` using five folds, shuffling, and seed 20260803. Groups were the 199 donors and strata were the 13 cell labels. Every cell from a donor was assigned to one test fold, giving 38–42 test donors per fold. Group overlap was checked programmatically before fitting, and every training and test partition was required to contain all classes. The five test folds form an exact, non-overlapping partition of all 462,034 cells.

Gene filtering, Seurat v3 ranking, vocabulary matching, representation construction, and classifier fitting were repeated inside each outer fold. Fold-specific gene sets therefore used only outer-training donors. Out-of-fold probabilities were reassembled in original-cell order only after all folds had completed. We report pooled out-of-fold metrics and the mean and sample standard deviation of fold metrics. This evaluates generalization to unseen donors within a shared cohort; it does not emulate transfer to a laboratory, technology, or label ontology absent from training [51].

5.8 Label-scarcity experiment

Within every outer donor fold, we subsampled the training cells to 10, 25, 50, 100, 250, or 500 labelled cells per class without replacement. Each budget was sampled with five seeds (20260811–20260815), independently within every class, while the outer-test donors and fold-specific representations remained unchanged. Logistic Regression and XGBoost were fitted to both representations, yielding 5 folds $\times$ 6 budgets $\times$ 5 seeds $\times$ 2 representations $\times$ 2 classifiers = 600 fits. For each fold and setting, seed-level results were averaged before the mean and standard deviation across outer folds were calculated. The design changes the amount of label supervision but not cohort or target ontology, and is therefore described as label scarcity rather than zero-shot transfer.

5.9 Temperature scaling

Post-hoc calibration was tested for Logistic Regression on each representation [32]. Within each outer fold, a second five-fold stratified group split was constructed from outer-training donors only, using seed $20260903 + f$, where f is the outer-fold index. The first candidate split whose classifier-fit and calibration partitions both contained every class was used. The classifier was fitted on the inner-fit donors. One positive temperature T was then selected on the inner-calibration donors by minimizing multiclass NLL,

$$T^* = \arg\min_{T>0} -\frac{1}{n}\sum_{i=1}^{n} \log \left[\mathrm{softmax}\left(\mathbf{z}_i/T\right)_{y_i}\right], \tag{1}$$

where $\mathbf{z}_i$ denotes the Logistic Regression logits. Optimization was performed over $\log T \in [-5, 5]$ with a bounded scalar optimizer. The fitted temperature was applied once to logits for the untouched outer-test donors. Uncalibrated and calibrated probabilities were evaluated on identical cells. Because T is a positive scalar, it preserves the ordering of logits and hence predicted classes.

5.10 Performance and calibration measures

Macro F1 was the unweighted mean of the class-specific F1 values and was selected to avoid dominance by abundant classes. Accuracy and balanced accuracy were also retained. For a test set of n cells, top-label equal-width ECE with ten bins was

$$\mathrm{ECE} = \sum_{b=1}^{10} \frac{|B_b|}{n} \left|\mathrm{acc}(B_b) - \mathrm{conf}(B_b)\right|, \tag{2}$$

where B_b contains cells whose maximum predicted probability falls in confidence interval b, $\mathrm{acc}(B_b)$ is the observed fraction correct, and $\mathrm{conf}(B_b)$ is mean maximum probability [32, 52]. The adaptive version replaced equal-width bins with ten equally populated rank groups. Class-wise ECE applied ten equal-width bins independently to the predicted probability of each class and averaged the class-specific errors. Because ECE depends on a binning choice and only summarizes selected aspects of calibration, it was not interpreted in isolation [33, 34].

For one-hot target vector $\mathbf{e}_{y_i}$ and predicted vector $\mathbf{p}_i$, the multiclass Brier score was

$$\text{Brier} = \frac{1}{n}\sum_{i=1}^{n}\|\mathbf{p}_i - \mathbf{e}_{y_i}\|_2^2, \tag{3}$$

and NLL was $-n^{-1}\sum_i \log p_{i,y_i}$. Both are strictly proper scoring rules that assess the complete probability vector, although their values reflect discrimination as well as calibration [35, 53]. Probabilities were clipped below at 10^{-12} and renormalized before scoring.

5.11 Bootstrap intervals and reliability diagrams

All intervals are percentile intervals from 5,000 bootstrap replicates. Within each primary atlas, we resampled the joint state comprising the true label and the predictions of the compared pipelines, preserving paired predictions and the original test-set size. Macro F1 was recalculated from the resulting confusion matrix. These cell-bootstrap intervals condition on the fitted models, training partition, and observed test-cell population; they do not represent independent biological replication or training-split variability.

For donor-held-out out-of-fold predictions, the primary uncertainty analysis sampled 199 donors with replacement and carried all cells of every selected donor together. Donor-specific confusion matrices and score contributions were combined using bootstrap multiplicities. We also calculated a cell-level comparison by multinomial resampling of the pooled confusion matrix. Paired differences used the same donor weights for both pipelines in each draw. The reported bootstrap probability that a difference is positive is the fraction of replicates above zero; it is a descriptive bootstrap quantity, not a frequentist P value.

Reliability diagrams used the ten equal-width top-label confidence bins defined above and displayed the number of cells in every bin. Within-atlas bin-accuracy intervals were Wilson 95% intervals. Donor-held-out bin intervals were percentile intervals from the donor bootstrap. An empty bin was retained with count zero and no accuracy estimate. This combination exposes whether an apparent calibration gap arises in a highly populated confidence region or from a small number of cells.

5.12 Computational audit and reproducibility

The complete workflow was executed in Python 3.11.10 with PyTorch 2.5.1 and CUDA 12.4, Scanpy 1.11.5, scikit-learn 1.9.0, and XGBoost 3.2.0. Random seeds, data identifiers, direct data download endpoints, checkpoint hashes, split assertions, preprocessing, metrics, bootstrap code, and figure generation are included with the submission. Every prediction archive records the model name, class names, test indices, true labels, and full probability matrix. Aggregate analyses assert exact cell coverage and alignment across pipelines before paired calculations. Raw public datasets and pretrained weights are not redistributed; scripts retrieve them from their original providers. No custom scGPT wrapper, end-to-end fine-tuning, gene-network analysis, or results derived from those earlier workflows is retained in the corrected numerical record.

6 Data availability

All datasets are publicly available through CZ CELL×GENE Discover. The collection page supplies publication and metadata context; the paired explorer link identifies the exact object analysed.

- TNBC breast cancer: https://cellxgene.cziscience.com/collections/ceef2841-5333-46ac-92ef-ccbe0c20fe55; exact dataset https://cellxgene.cziscience.com/e/6f9de485-58cd-4342-bfc4-b3d3dd223aa8.cxg/.
- Indonesia PBMC: https://cellxgene.cziscience.com/collections/d1e0e64d-6d2a-4a3e-b7f4-43ed909a9d9c; exact dataset https://cellxgene.cziscience.com/e/ca7d95ac-53aa-4a94-8c76-300c608ce6a0.cxg/.
- Brain atlas: https://cellxgene.cziscience.com/collections/0986e4cd-7a58-405d-9b91-4b199bb4124e; exact dataset https://cellxgene.cziscience.com/e/e2a00644-0a48-4815-ae87-d562045114f5.cxg/.
- Multi-tissue TME: https://cellxgene.cziscience.com/collections/3f7c572c-cd73-4b51-a313-207c7f20f188; exact dataset https://cellxgene.cziscience.com/e/b617ee1b-f8c8-4de9-b82b-e803ab93550d.cxg/.
- Human pancreas: https://cellxgene.cziscience.com/collections/0a77d4c0-d5d0-40f0-aa1a-5e1429bcbd7e; exact dataset https://cellxgene.cziscience.com/e/3294d050-6eeb-4a00-b24c-71aacc9b777f.cxg/.
- Pig pancreas: https://cellxgene.cziscience.com/collections/0a77d4c0-d5d0-40f0-aa1a-5e1429bcbd7e; exact dataset https://cellxgene.cziscience.com/e/db4a9ed2-e994-40c1-b7ec-4091fdf7b6c1.cxg/.

The released whole-human scGPT checkpoint is available from the scGPT authors at https://github.com/bowang-lab/scGPT. Data and checkpoint identities are independently recorded by stable dataset UUIDs and SHA-256 hashes in the computational audit. Summary tables, reliability-bin data, bootstrap results, and executable analysis code are included in the submission package and the versioned public repository snapshot at https://github.com/alirezakhosravii/scGPTvsClassical/tree/scientific-reports-revision-2026-08-09. Raw H5AD objects, per-cell prediction archives, and pretrained weights are not duplicated because of their size and existing public availability.

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Declarations

Ethics approval and consent to participate. Not applicable. This study used only publicly available, de-identified single-cell datasets released under the original studies' ethics approvals.

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